The Warehouse Inventory Balancer

Week 1 Assignment - Problem 4 | Java

Problem Summary

Calculate both section totals, report Balanced or Not Balanced, and find the highest individual quantity across both arrays. Ties retain the first location: Section A is scanned before Section B.

Java Solution

```
public class Main {
    static void analyzeInventory(int[] sectionA, int[] sectionB) {
        if (sectionA.length != sectionB.length) {
            System.out.println("Sections must have equal length");
            return;
        }
        int totalA = 0, totalB = 0;
        int max = Integer.MIN_VALUE, maxIndex = 0;
        String maxSection = "A";
        for (int i = 0; i < sectionA.length; i++) {
            totalA += sectionA[i];
            if (sectionA[i] > max) {
                max = sectionA[i]; maxSection = "A"; maxIndex = i;
            }
        }
        for (int i = 0; i < sectionB.length; i++) {
            totalB += sectionB[i];
            if (sectionB[i] > max) {
                max = sectionB[i]; maxSection = "B"; maxIndex = i;
            }
        }
        String status = totalA == totalB ? "Balanced" : "Not Balanced";
        System.out.printf("Section A Total: %d | Section B Total: %d | Status: %s | ",
            totalA, totalB, status);
        System.out.printf("Highest Quantity: %d (Section %s, Item %d)%n",
            max, maxSection, maxIndex + 1);
    }
    public static void main(String[] args) {
        analyzeInventory(new int[]{20,15,30}, new int[]{25,10,30});
    }
}
```

Sample Output

```
Section A Total: 65 | Section B Total: 65 | Status: Balanced | Highest Quantity: 30 (Section A, Item 3)
```

Complexity / Key Notes

Time: O(n) for each pair of arrays of equal length. Extra space: O(1). Strict comparisons preserve the first occurrence of a tied maximum.